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Breast Cancer: Epidemiology Therapeutic and Diagnostic Approaches

Abstract:

¹⁰⁷ Breast cancer remains to be one of the top reasons for cancer-linked fatalities in women every year however, breast cancer also has specific epidemiologies and various forms of biological behaviour. Based on the progression of knowledge surrounding the different molecular subtypes, there should be an even greater understanding and accuracy within care of breast tumor. In an effort to obtain an greater awareness of tumorigenesis and the processes of progression of breast cancer, there has been an increased focus on research related to many different aspects, including tumor stem cells, intra-tumoral microbiota, and circadian rhythms. In addition, advancements in technology, especially through artificial intelligence, will lead to improved detection accuracy of breast cancer. New ideas for treatment and new medications for breast cancer continue to change the way we think about, and treat, breast cancer (personalised medicine). Based on the existing empirical research data, it is apparent that diagnosis and treatment Requires personalised guided by an personal chance and subtype of breast cancer, this would ensure the realization of all anticipated advantages of consistency oncology in breast cancer. The following are the four main topics included in the present review; (a) Evaluation and ¹prediction of the risk of breast cancer. (b) Information based on evidence regarding effective preventive strategies and ways of improving lifestyle to minimize chances of acquiring breast cancer. (c) Knowledge about breast biology would help in developing new strategies for prevention of breast cancer. (d) a report summarizing recognized procedures that can be applied to prevent any potential problems as well as what third party resources will be used to implement those procedures.

Introduction

The most widely detected ³³ kind of cancer in women today is breast cancer, and as society becomes more aware of breast cancer, becomes more knowledgeable about the disease, and has more access to breast imaging technology, the rate at which female are detected with breast cancer earlier in their lives will continue to rise. Breast cancer kills women. Breast cancer is the most frequently diagnosed type of cancer in women. The study of

mechanisms of breast cancer tumorigenesis and progression has been of paramount importance in scientific research, with many researchers addressing these mechanisms through multiple lenses. Advanced innovation, in particular with respect to the use of artificial intelligence, has greatly developed the accuracy of detecting and diagnosing breast cancer. In addition, despite early detection and treatment advances that have improved in recent years, the underlying origins and vulnerability factors of the illness are complex and should have additional attention, as we aim to boost our knowledge of breast cancer at the molecular level. Multiple factors can lead to the development of tumors associated with breast cancer. The literature indicates that these factors are multifactorial in nature (i.e., genetic, environmental, lifestyle factors), and that the interplay of these three different kinds of factors is complex in their contribution to this process (Xiong et al., 2025). Other than the additional variables that will affect the growth of tumors like the tumor stem cell phenomenon, the presence of bacteria in tumors, and circadian rhythms, there is a need to study the biology of the other parts as well to establish other potential therapy sites for application in the clinic. Breast cancer stands as the most frequently occurring category of cancer in female patients and is consistently one of the most prevalent forms of any kind of cancer globally. There are an estimated 5.2 million women alive five years following a breast cancer diagnosis (Justo et al., 2013).

Global incidence and prevalence

According to reports from the World Health Organization (WHO), breast cancer rates are the highest of any cancer type in Africa, which ranks first worldwide compared to North America, Europe, Asia and Australia for the number of cases reported in 2008. Sub-Saharan Africa has among the highest incidence rates of breast cancer compared to all other cancers overall and the incidence of breast cancer varies greatly among continents; Eastern Africa has the lowest rate of breast cancer incidence worldwide, which is 30.4 per 100,000 female population each year" (WHO, 2008) , while Southern Africa has the highest rate of 38.9 per 100,000 women per year. There is the potential that women of colour living in Southern Africa may have some of the highest breast cancer incidences due to the large percentage of women of colour in Southern Africa compared to other regions of the world. France has a 5-year and a 10-year survival rate

of 88 % and 78% respectively, making them one of the regions with the best survival rates in Western Europe (Sancho-Garnier & Colonna, 2019).

²³ Breast cancer has become the leading cancer diagnosis for women in Latin America because of the growing incidences and mortality rates due to breast cancer in Latin America. Each year in Latin America and the Caribbean there are about 114,900 cases of breast cancer and around 37,000 deaths resulting from the disease (Justo et al., 2013).

The frequency of breast cancer cases for women in Western Europe is five times higher than for women in West Africa with an estimated 450,000 women newly diagnosed and 140,000 deaths due to breast cancer in 2008 (Abdulrahman & Rahman, 2012).

Breast cancer incidences are higher in Pakistan than in many other Asian countries [4]. Additionally, breast cancer deaths are also prevalent because of late diagnoses or patients being referred after the diagnosis to specialist treatment facilities (Zaheer, Shah, Maqbool & Soomro, 2019). Globally, there are huge incidences of cancer-related deaths in many nations all around the world, including countries in Asia where cancer-related deaths have risen dramatically throughout the continent. In this regard, Asia also shows high overall rates. During 2020, records showed nearly 19.3 million newly reported malignancy cases diagnosed globally (including estimates from Pakistan) and 10 million cancer deaths (World Health Organization, 2021). Also, according to WHO, amongst new and existing (i.e. total) cancers, breast cancer has an overall incidence rate of 47.8 percent. Collecting cancer-related epidemiological effects for prospective data over time, especially in developing countries, will present significant challenges in terms of data collection as seen here within developing countries, particularly Pakistan in 2020, which had a population estimate of 220 million people (Siddiqui, Mehmood, & Khan, 2024).

Risk Factors

Genetic Factor

Hereditary breast cancer is found in estimated 5-10% of breast cancer diagnoses, and this is caused by mutations in the breast cancer tendency genes BRCA1 (breast cancer 1) and BRCA2 (breast cancer 2). Researchers have estimated that carriers of BRCA1 mutations have a risk of developing breast cancer ranging between 40% and 87% and

that the risk for carriers of BRCA2 mutations range between 27% and 84% (Kuchenbaecker et al 2017). Both BRCA1 and BRCA2 had very high risk estimates when compared to other genes, with BRCA1 and BRCA2 forming a tumour-suppressor gene pair through homologous recombination, known as DNA dual strand damage restoration. When BRCA1 and BRCA2 become mutated, they can no longer function as effective tumor suppressors. The rarity of BRCA1 and BRCA2 mutations in the population has rendered it difficult to assess historical penetrance accurately in retrospective studies. A mutation screening programme was carried out among women selected based on including early stage of identification as well as familial record of tumor (King, Marks & Mandell, 2003).

Beyond BRCA1/2, the growth catalog of moderate-to-significant penetrance genes — including PALB2, CHEK2, ATM (as the evidence suggests), and CDH1 — confer elevated risk and are increasingly assessed through multi-gene panel testing. Research papers on extensive genetic association studies have described more than 300 individual SNPs (Single Nucleotide Polymorphisms) connected to an average increase in risk for evolving breast cancer. This research has also contributed significantly towards developing polygenic threat rating systems for the estimation of likelihood breast cancer, as stated by Michailidou et.al., 2015.

The likelihood that someone will develop breast cancer can be predicted using family history, meaning that family history is a good clue about your likelihood of developing cancer. If you have a blood relative detected with breast cancer (e.g., a parent, child, sibling), your risk of developing breast cancer is twice that of someone without such a blood relative. In contrast, if you've had blood relatives (parent, child, sibling) detected with breast tumor, your relative risk of acquiring breast cancer is three or more times that of a female without blood relatives diagnosed with breast cancer. Family history of breast cancer risk is different than hereditary genetic risk factors from that parent (cancer 2001).

Hormonal factors

Cumulative exposure to oestrogen has been started as an separate predictor of building hormone receptor positive breast cancer. Eight of ten studies show that people with increased levels of serum estradiol (sE2) (coconut oil) are at an increased risk for developing breast cancer. The pooled relative risk estimate for elevated sE2 levels is 2.3 (95% CI: 1.6-3.2) [Key & Verkasalo, 1999]. There is a significant amount of

evidence linking elevated sE2 levels ¹⁴ in post-menopausal women to increased risk of acquiring breast cancer. As oestradiol (and many other hormones) behave in this way ³ plays a role in breast cancer, they most probably exert their influence through controlling the rate of proliferation (rate of division) of cells within breast epithelium. High mitotic rates in breast epithelial cells can increase the probability that genetic alterations arise and get copied before being repaired, thus, high mitotic rates also increase the likelihood that early-stage cancer will develop.

Substances that alter hormonal levels such as certain medications may dramatically affect how the individual responds to particular environmental and other influences. For example, Combined Estrogen and Progesterone hormone replacement therapy in postmenopausal females is related with increased incidence for developing breast cancer by approximately 26% (Women's Health Initiative: NIH, [George, 2003]). Research regarding combined oral contraceptive use and IV also supports this theory, regardless of level of data available, with lower level of association when compared to combined hormone replacement (Cheelbow et al, 2003).

Female patients suffering from breast cancer average fewer children than women without ³ breast cancer. A woman's associated chance of breast cancer reduces by 4.3% for each additional child she breastfeeds, from 2.9% RR to 5.8% RR; a woman's associated chance of breast tumor decreases by 7.0% for for each subsequent child she gives birth to, from 5.0% RR to 9.0% RR. This decrease in risk can be attributed to the endocrine effects of both the suppression of ovulation and the change in the hormone profile associated with lactation. Additionally, if all women in developed countries were to give birth and breastfeed for the average lifetime duration that was the norm in developing countries until recently, there is a projected 50% decrease (from 6.3% to 2.7% per 100 women at age 70) in breast cancer incidence. Two-thirds of this reduction in incidence is believed to be attributed to breastfeeding (Kerr, 2002).

Lifestyle factors:

The percentage of breast cancer is comprised of multiple factors related to lifestyle, the majority of which can be modified, and they all contribute to the large share of the linked

susceptibility connected to breast cancer. For example, postmenopausal women gain weight (or fat) to increase the amount of circulating estradiol through ⁹¹the conversion of androgens to estradiol via the aromatase enzyme present in the fat (adipose) tissue. As shown in various studies, estrogen is the primary hormone that causes breast cancer. Therefore, obesity and adiposity reduce a woman's breast cancer risk with respect to her risk compared to a woman of normal weight; by 20-40% greater than the risk of a comparable woman of normal weight. Studies reveal that there exists a link between the rising incidences of breast tumor amongst postmenopausal women by having a BMI of 5 kg/m² increments. The rise might be seen as minimal but clinically it has significance, (see Renehan et al., 2008).

Alcohol consumption has a dose-dependent connection with breast cancer risk; even moderate intake (one drink per day) expands risk by nearly 7–10%, likely through acetaldehyde-mediated DNA damage and hormonal effects. Sedentariness presents the largest ²⁰risk factor for developing breast cancer in females. The national cancer authority reports that females who exercise daily (i.e. low to moderate levels of intensity), decrease their threat of developing breast cancer by 10 to 20% as compared to females who are more sedentary. While the body metabolizes alcohol, one specific by-product of this metabolism, acetaldehyde, is considered to be a carcinogen, as it actively participates in damaging DNA. Alcohol increases the concentration of hormones (including estradiol and estrone) circulating in a woman's body. Studies have shown that alcohol consumption is linked with an raised threat of ¹⁰⁵estrogen hormonal positive breast cancer in women, implying that the hormone response involved in breast cancer is a key driver of its development (³⁰Allen et al., 2009).

The threat of evolving breast cancer at an earlier age than expected (during a critical stage of breast development) continues to rise among women treated using radiotherapy to the chest with no apparent plateau. The same phenomenon has been observed among the bomb survivors from Hiroshima and Nagasaki and has important implications for the area of treatment research. After the atomic radiation exposure, the number of breast cancer and other diseases increased drastically. Patients with Hodgkin lymphoma receiving thoracic radiation therapy can also experience the same results as patients undergoing repetitive diagnostic imaging (Henderson et al., 2010).

Age distribution

⁵ Breast cancer threat is related to aging. The median age of a breast cancer patient is currently ⁴53 years old and breast cancer is prevalent in the United States (49 years) and female who are 50.1 years of age are the most common in this particular study, but studies show that women in developing countries experience breast cancer at much younger ages (Chopra et al., 2014).

Mortality trends

By the year 2022, breast cancer will be liable for an estimated 670,000 deaths worldwide; it will also rank fifth among cancer deaths worldwide. The CISNET Breast Cancer Working Group published their results of the role that both adjuvant therapy and mammographic screening played in lowering breast cancer mortality rates (Ferlay et al., 2021). From 1975 - 1990, the death rate for women from breast tumor increased from 48.3 to 49.7 deaths per 100,000 women. By 2000, the breast tumor death rate had declined by 24% (38.0 per 100,000 women) from its ⁹²rate of 49.7 deaths per 100,000 women in 1990. Other countries have had similar reductions in deaths due to breast cancer. The feasible cause for the reduction in breast cancer deaths in both countries include As such, two factors were noted in the findings for LCIS. First, ²⁷initial diagnosis of breast cancer became feasible through the use of mammography. Second, many women were detected with very initial phase breast cancer.

While over seventy percent of breast cancer had been formerly detected at diagnosis stages I/II within the Countries of Higher Income, only twenty percent to fifty percent of breast cancer patients have previously been detected in those stages in general limited- and Middle-Income countries (Unger-Saldaña, 2014). Mortality remains high and continues to rise in many LMICs, including features of sub-Saharan Africa and South Asia, where five-year survival rates may be as decreased as 40–50% compared to over 90% in developed nations. These disparities mirror variations in stage at diagnosis, access to comprehensive care, and the availability of molecularly targeted treatments (Worsham et al., 2007).

An additional importance has also been assigned on the continuing accuracy of research findings. In 2012, breast cancer continues as the most frequently occurring tumor form comprising 13.8% of all recorded cases globally. In 2012, breast cancer ranking after lung cancer as the primary reason of death from cancer among female in China, India and Russia (Unger-Saldaña, 2014). Since there are multiple variables contributing to an

⁹risen incidence of breast cancer, it is difficult to link increasing cases of breast cancer to just one reason. For example, inc. in breast cancer from 2005 to 2015 were at 33% totals of cases. The demographics of the population provide an explanation for the increase, with twelve point 6% of new cases identified due to population growth; sixteen point 4% of new cases result from age; four point 1% resulted from age specific cases. Total mortality cases in the European Union decreased in 2012 by 9,100 still there was an overall rise in total female breast cancer mortality 21.3% from 439,800 total to 533,600 total from 2005 through 2015 (Azamjah et al., 2019).

Pathophysiology and Classification

Development of breast cancer:

Breast tumor typically develops as a gradual process, starting with inherited and epigenetic alterations over time, ultimately leading to gradually accumulating genetic and anatomical errors. The mainly basic ¹⁰¹feature of cancerous cells is their ability to continue growing and dividing forever, without limit or constraint. In a normal biological system, the growth and release of growth factors regulate the number of cells present in a given tissue, which allows that tissue to support a functional architecture as well as supporting the normal function of the cells it contains. This being the case, cancer cells, by deregulating these signals, become masters of their own destinies, which merits closer examination. It must also be recognized that the traditional model for the progression from normal epithelium to hyperplasia, abnormal atypical hyperplasia, in situ carcinoma, and ultimately into aggressive cancer has been significantly revised by data obtained from molecular profiling studies that have elucidated unique functions related to each subtype of breast cancer. (Hanahan & Weinberg, 2011)

Tumors can develop as a result of genetic changes, but they can also develop as a result of changes to the epigenome. By silencing tumor suppressor genes and activating pathways that promote cell growth, epigenetic changes can contribute to the process of oncogenesis. Breast cancers are very diverse in nature, but most cases originate in the ducts of the breast. The best known genetic alteration associated with lobular breast cancer (ILC) is loss of E-cadherin; however, we have also identified ILC-specific

mutations associated with PTEN, TBX3, and FOXA1. Among all the breast cancer subtypes, ILC has the highest (PTEN-associated) increase in phosphorylated AKT. A subgroup of FOXA1 mutations showed increased FOXA1 activity and expression in a spatially clustered manner. GATA3 mutations and high expression contribute to the lower incidence of invasive ductal carcinoma (IDC) of the luminal A phenotype as compared to invasive lobular carcinoma (ILC), suggesting that the mechanisms by which the estrogen receptor (ER) is regulated may differ among ILC and IDC; therefore, different mechanisms are likely to regulate ER-positive ILC and IDC (Ciriello et al. 2015).

The cancer microenvironment is now known to have a significant impact on the development of tumors and their response to treatment. Studies that have recently been published have demonstrated that the gene expression and epigenetic alterations for cells making up the microenvironment differ significantly throughout tumor evolution. Moreover, these differences may be identified and used as biomarkers or targets for therapy. The breast microenvironment contains extracellular matrix (ECM) and an extensive range of different kinds of cells comprising the stroma, including Endothelial cells, immune cells, fibroblasts and adipocytes (Place, Jin Huh, & Polyak, 2011).

⁴⁹Types of breast cancer

Ductal carcinoma

Invasive Ductal Carcinoma (IDC) - also called Invasive Breast Carcinoma and/or Infiltrating Carcinoma, is thought to develop from the milk ducts of the breast (as suggested by the data). IDC typically grows and spreads by breaking through the duct wall (as suggested by the evidence) and infiltrating into adjacent breast tissue. IDC can metastasize outside the breast via two distinct mechanisms - via the blood or lymphatic systems. Entirely, IDC accounts for about 80% of all invasive breast cancer cases (8 out of 10). Likewise, IDC is also the most frequent malignancy in men as well. However, this is not the only reason IDC is referred to as "Ductal Carcinoma"; there are also other reasons, such as the fact that Ductal Carcinoma can develop from precancerous tissue that has been detected and diagnosed at early-stage. By contrast to this, 1 out of every 5 new breast cancers is diagnosed before they become invasive (Stage I-IV).

DCIS (Ductal Carcinoma In Situ) is one type of cancer that is non-invasive, it's completely contained within the nipple ducts (or ductal basement), and it can eventually form invasive ductal carcinoma (IDC). The majority of DCIS lesions have been identified as micro-calcifications on mammograms and are currently estimated to account for approximately 20-25% of breast cancers detected through screening. Most of these lesions (DCIS) typically do not alter or progress towards becoming invasive, however, it is still a considerable hurdle to anticipate which patients will proceed from DCIS to an invasive breast carcinoma (Zhou et al. 2017). In general, the prognosis for DCIS patients is quite good, but some will go on to develop invasive cancer or die of metastatic disease. The ability to segregate patients at increased risk for eventual recurrence or even an enhanced risk of dying from breast carcinoma is likely to alter a number of patients therapeutic modalities. There is evidence to support that many women could be spared treatment (radiation therapy and/or mastectomy) if we were better able to assess their likelihood of progressing toward penetrating breast cancer (Morrow et al., 2016).

Lobular Carcinoma

Breast cancer is a heterogeneous disease due to the histological variability found within breast cancer types. Lobular Carcinoma (LC) is a subtype of breast cancer which comprises 8-14% of all breast cancers. Data from a recent epidemiologic investigation have provided some initial insight into the possible causes for this particular type of breast cancer, although the specific causal factors remain undefined.

Lobular and ductal carcinoma differ in morphology. Lobular carcinoma appears as small round cells that are bland in appearance and possess very little cytoplasm when looked at under a microscope. These small, round cells infiltrate the stroma (the connecting tissue in the breast) as individual cells in a line, and also surround normal breast tissue in a very specific way. In most instances, lobular carcinoma infiltration does not result in damage to the anatomy of the breast. Due to the unique growth pattern of lobular carcinomas and their biology, lobular carcinomas do not often create easily palpable masses or can be visually identified using mammography; thus many lobular carcinomas are difficult to locate and diagnose using these methods. The literature suggests, tentatively, that this can make early diagnosis challenging and breast conservation frameworks more difficult.(Arpino, Bardou, Clark, & Elledge, 2004)

⁴¹ Lobular carcinoma in situ (LCIS) has slowly evolved over time. Historically, because LCIS had been considered a marker of moderately elevated risk for breast carcinoma; as opposed to a precursor lesion for invasive carcinoma; the reason being that both sides of the breast have the same increase chance of developing breast carcinoma. There may be a 50:50 chance that a patient who has developed an invasive carcinoma will have developed it more than 10 years after being diagnosed with LCIS. Most invasive carcinomas will have a ductal type of histology. Screening mammograms have increased the number of added incidental diagnoses of LCIS.(Hwang et al., 2004)

Additionally, the use of more specific histological assessments shows that individuals with LCIS usually present an adjacent tumor either with ILC or DC. Of the LCIS samples examined, there was solid evidence to suggest that lymphocyte infiltrate (E-cadherin) and other catenins (α -, β -, γ - α -catenins) were lost at the same time in the LCIS sample and also to some degree this was noted in the invasive component proximal to the LCIS. This suggested that ¹⁰⁴ the loss of E-cadherin and α -, β -, and γ -catenin would help develop lobular carcinoma in situ that would subsequently become invasive lobular breast carcinoma.

Importantly, other events, beyond the inactivation of E-cadherin, are believed to have a role in transitioning LCIS to ILC (de Leeuw et al., 1997). The highest rate of incident ILC occurs in White females, but the majority of female deaths due to ILC are among Black females (Pramod et al., 2021).

Molecular subtypes

Hormone receptor Positive Breast Cancer:

There are specific proteins present on or within the tumor cells, i.e., progesterone and/or estrogen receptors and, therefore, are referred to as ⁹⁵ hormone receptor positive breast cancer. It is possible to suggest that these receptors function similarly to locks, with circulating hormones acting as the keys. One could contend that when progesterone or estrogen attaches to these receptors, signals are sent that encourage the growth and division of cancer cells(Goldberg et al., 2021). You can usually halt or completely stop

that escalation by removing or blocking the hormones. While this may be true, this insight, first explored therapeutically in the late 19th century when Scottish physician George Beatson ¹⁰⁹observed that removing the ovaries of women with breast cancer caused tumor regression, became the foundation of highly efficient specific therapy strategies in the history of cancer medicine (Goldhirsch et al., 2013).

⁵³Breast cancer is the most common type of diagnosed breast cancer when HR+ is taken into account. Furthermore, this ¹is the most dominant molecular subtype of breast cancer worldwide and accounts for almost 69% of cases diagnosed in the United States. It could be suggested that despite this, hR+ tumors as a rule carry more favorable curtailed-term prognosis compared to HER2-favorable and triple-problematic subtypes, though they are associated with late relapse and persistent recurrence risk extending beyond 10 years from diagnosis (Sparano et al., 2018).

⁵¹HER2 Positive Breast Cancer

Breast tumor is the highly detected type of tumor in women globally and is the result of abnormal tissue expansion that ultimately guide to malignant cell expansion. Recent statistical data suggest that ⁹breast cancer constitutes for 11% of all diagnosed cancer cases globally annually and continues standing as a major cause of loss of life for female patients (Asif, Sultana, Ahmed, Akhtar, & Tariq, 2016)

⁹⁸HER2 is a protein that normally helps the breast cells communicate with one another to grow and reproduce correctly. HER2 also causes abnormal growth of breast cancer cells when it signals incorrectly, resulting in the gene creating multiple copies of itself. Together, HER-2-positive (HER2) breast cancers are classified as aggressive breast cancers with a higher rate of growth than other subtypes and greater likelihood of metastasis. Researchers are studying how to identify which ⁷patients with HER-2-positive breast cancer may require anthracycline-based chemotherapy (more specifically, doxorubicin) and which patients can safely avoid anthracycline therapy and its associated toxicities. (Engel & Kaklamani, 2007)

In breast cancer for women, HER2 is one of the requirements for use of drug therapy. According to Yates and coworkers, efficacious treatments use either the monoclonal antibody, trastuzumab (which is HER2 targeted), or the small molecule signaling enzyme targeting agent, lapatinib (which is HER1 and HER2 targeted). In women with advanced breast cancer, trastuzumab-DM1 (an antibody-drug conjugate) and

pertuzumab (an inhibitor of HER dimerization), have demonstrated superior effectiveness over traditional trastuzumab therapy. There are five currently in adjuvant studies evaluating the combined effectiveness of trastuzumab-based, as well as other, general ¹¹⁸ treatment for women with early-stage breast cancer: lapatinib [anti-VEGF], bevacizumab (anti-VEGF), neratinib ¹⁵ (dual inhibitors of HER1 and HER2), GP2 and AE37 (peptide vaccines). (Arteaga et al., 2012)

Triple negative breast Cancer

In 2007 there were approximately 465,000 fatalities out of 3 million breast cancer diagnoses, placing it as the initial tumor diagnosis in female, as well as the main factor for breast cancer-associated deaths.

Indicators ¹² of triple-negative breast cancer may also be identified through ²⁵ the absence of ER, PR, and/or HER2 as measured by IHC (Cleator et al. 2007). The three primary subtypes of breast cancer are luminal or hormonally receptor positive, HER2 amplified/ER negative, and basal-like (Irvin Jr. & Carey, 2008).

TNBC comprises ⁵² 10-20% of breast cancer cases, is typically diagnosed in younger African-American women, and is usually diagnosed with more advanced tumor grades or mitotic counts and at later clinical stages when compared to the majority of other cases of breast cancer. ⁸² The locoregional treatment options for TNBC are the same as ⁴² for invasive breast cancer, and may consist of a mastectomy (with or without adjuvant radiotherapy) or a lumpectomy (+ or - adjuvant radiotherapy). It has been proposed that the lack of drug-targeting receptors on TNBC cells limits the treatment options available for improving outcomes, and that chemotherapy remains the only ¹⁵⁰ therapeutic option for patients with TNBC. The literature ²² provides evidence that chemotherapy has been the most effective intervention in treating Triple Negative Breast Cancer (TNBC) by virtue of the high proportion of patients with the disease achieving pCR in response to neoadjuvant chemotherapy. Neoadjuvant chemotherapy has been validated even more in breast conserving surgery by confirming the pCR status, (see Kumar & Aggarwal, 2016).

Sign and symptoms

Thus, the symptoms that may present in an individual ⁸¹ with breast cancer will vary widely depending on the subtype of breast cancer diagnosed and how advanced the disease is at the time it is diagnosed. for example, there are typically no symptoms

present in early stage breast cancer, so it's critical to have regular breast cancer screenings done to detect these diseases.

Common sign and symptoms include

Lumps in the breast, such as:

- Lumps in or around the breast area.
- Abnormal lumps underneath your arm.
- Thick or hard areas (or lumps) for Breast tissue or lumps underneath your arm along with breast changes in size or shape.
- Nipple alterations or discharges may include:

•Fluid or discharge other than breast milk.

•Changes in how the nipple appears; for example, the nipple becomes changed, twists, flat. Changing in skin such as:

- the skin around the nipple may become red
- irritation and itching around the nipple
- Generalized swelling throughout the breast without a discernible lump
- skin rashes
- Dimples or puckering .Additionally, advanced-stage metastatic breast cancer might lead to more specific symptoms linked to distant organs (Koo et al., 2017).

Cancerous ⁹breast tissue may spread to other parts of the body and cause other symptoms, including back pain, bone pain or fractures, trouble breathing, a chronic dry cough, abdominal pain and swelling or yellowish skin (jaundice), headaches that are severe & persistent, seizures and/or problem with your vision (Cardoso et al., 2018).

¹¹³Stages of breast cancer

The clinical prognostic stage is used to identify a breast cancer stage for the diagnosed patient. The clinical prognostic stage includes total health history, results from physical examination, imaging test results (including mammogram and ultrasound exam of

lymph nodes) as well as biopsy results from tissue taken at the breast's original site. The clinical prognostic stage is based on the patient's present disease state while the pathological prognostic stage will take only into account the patient's detailed health history, physical exam findings, results of ⁷⁵imaging tests such as mammography and/or ultrasound, and pathological findings from biopsy. The pathological prognostic stage (a valid classification only for breast cancer patients who have had surgery) consists entirely of clinical patient data; collected and assembled before the patient underwent surgery (including biomarker data) from biopsied tissue from the removed breast and lymph nodes. Each prognostic stage is assigned a number from 0-IV (4) and is calculated according to patient tumor size, range of lymph node engagement, and any evidence of distal tumor spread, so the combination of all three classifications (TNM) represents the labeling of breast cancer; in addition to tissue examination grade, surgical margins, and any associated biomarker status (HR and HER2) for that tumor. In some cases, results of a multigene test for breast cancer can be used to assist with staging ³⁷(Amin et al, 2017).

Stage 0 (Ductal Carcinoma In Situ) breast cancer is non-invasive. For a breast cancer case to be considered non-invasive there must be abnormal breast tissue cells. These abnormal cells originate and remain in that area; hence, they are considered metastatic if they spread beyond their original site. Non-invasive breast cancer can have any histology grade; non-invasive breast cancer can also have any biomarker status (HER2, ER, PR).

Stage I breast cancer consists of small breast tumors that may exist in the breast tissue itself, as well as potentially involve lymph nodes in close proximity. Breast Cancer Stage I consists of two portions - Stage IA and Stage IB. Determination of whether you have breast cancer Stage IA / IB requires an evaluation of ⁴³the size of the breast cancer and whether there are cancer-positive lymph nodes.

⁹³The staging of Stage II Breast Cancer shows that breast cancer is present only within the breast tissue ²⁴or it can involve lymph nodes near the breast region or both within the breast tissue and lymph nodes. Stage II breast cancer has also been categorized into two subtypes; IIA and IIB. This classification of Stage II breast cancer has been done based on the size of the tumor present within the breast along with its invasion into the lymph nodes..

Having a stage III breast cancer diagnosis means your cancer has involved one or more of these areas: the nodal tissue near your breast area, ¹²³the skin above the breast, and the chest wall itself.

If you ⁹⁹have been diagnosed with Stage IV Breast cancer, it implies ¹⁶that the original primary tumor which is located in your breast tissue and ¹⁰⁸lymph nodes adjacent to the breast tissue has led to locoregional disease or cancer. Such type of cancer usually has metastasized to distant sites like bones, liver, lungs, and other organs of the body. Metastatic ¹⁰⁸breast cancer refers to any tumor stage of breast cancer at a distant site from the original location of the tumor (Zubair, Wang, & Ali, 2021).

Complications

At the heart of this inquiry is breast cancer and its treatments are associated with significant narrow- and long-term complications. Disease-affiliated complications contain lymphedema — chronic arm swelling resulting from axillary lymph node dissection, more precisely, or radiation-induced lymphatic obstruction — affecting up to 30% of patients following surgical axillary clearance (Armer & Stewart, 2005). ⁵⁶Metastatic breast cancer, also known as Stage IV breast cancer, has a variety of complications based ¹³on the organ(s) affected by metastasis from the primary tumor. The most frequent ¹⁹organs affected by metastatic breast cancer are bone, lung, liver, brain, and are responsible for the symptoms associated with this disease such as fatigue and nausea, as well as the loss of appetite and loss of weight. Most complications can be treated, however there is a varied chance of survival based upon your location, size of the tumour(s), and how many tumours you have. Treatment is available for stage 4 breast cancer, but no cure exists for this stage. Approximately ⁷⁶70% of patients with stage IV breast cancer will have experienced metastases initially in the bones, before they have metastasized to other organ(s). ² The presence of cancerous tissue in bone(s) affects a process known as "bone remodelling," which is the continuous breaking down and replacement (with new bone cells) of bone tissue (Lim & Hortobagyi, 2016).

Chemotherapy can have both short and long term side effects. temporary outcomes typically happen while a patient is being treated, and usually go away by the time treatment is finished, whereas some long term effects may not show up until many years have passed after treatment and continue showing signs for a much longer period of

time(Partridge, Burstein, & Winer). Rarely will patients show long term effects, such as secondary leukemia until many years have passed after completing chemotherapy (e.g., 5 to 15 years), as well as premature ovarian failure in subsets of patients. ⁸ There is little evidence but the majority of patients with chemotherapy do not experience these rare long term effects and therefore should not be used as a decisive factor in making decisions regarding adding adjuvant therapy when the absolute risk of occurrence is very small. Patients can also have experienced menopausal symptoms, experience osteoporosis, and develop chronic musculoskeletal pain due to adjuvant endocrine therapies that have significant clinical implications(Partridge et al.).

Diagnostic Approaches

Clinical Examination

CBE is designed to identify palpable breast cancers before they are far along in the growth process so that treatments will be more successful to use, and many more treatment options will be available to the cancer patient than would be available for late stage disease. In past research, there were many studies done looking at the amount of breast cancers detected only by physical examination, or clinical breast examination (CBE). Most of these studies were conducted in previous years and mammography had not developed as much then as it has now. In the studies that have been conducted recently, three found between 4.6% and 5.7% of cancers could only be found with CBE while another found 10.7%. In performing a CBE, a triage physician looks for evidence of advanced breast cancer and thus looks for differences or defects in shape such as asymmetry, differences in color, texture or temperature of the breast, and large differences in the venous pattern from the two breasts when comparing them. Any rashes, discolouration, lumps, swelling or discharge from the nipple must be subject to particularly close inspection. Signs of advanced breast cancer can be identified based on a standardized assessment method using the acronym BREAST which includes different breast masses, large differences between the two breasts, signs of edema, tenderness, scaliness etc. (McDonald, et al., 2004).

Imaging techniques:

Mammography

The process of Diagnosing breast cancer has now been identified as an important imaging test. The method of diagnosis uses the non-invasive technique of mammography along with low dose x-ray (absorption) to produce very high quality images of the interior of the breast. The process of mammography operates on the basis of various types of breast tissue absorb x-rays differently (degree of difference) due to how much compression is applied between two plates of a mammography machine; the breast is heavily compressed, and therefore, the amount of radiation (through the use of a decreased radiation dose) that will be projected through the breast will vary based on the types of breast tissue (i.e., fatty and cancerous) that are in the breast; all of these types will absorb the x-rays differently to create the image; the actual x-ray (digital detector) is recorded from the x-ray (digital detector) after it has passed through the breast tissue; the resulting image produced by the digital detector produces a 2D representation of the chest area. Mammography has undergone significant improvements over the last several years. ⁵⁴ Digital mammography (DM) has replaced previous film-screen mammography (FM; however), DM has produced improved quality images while using less radiation than FM. There is also less chance of false positives in DBT (Digital Breast Tomosynthesis; or, or 3D mammograms) than would otherwise occur during a routine screening for breast cancer. ABUS (Automated Breast Ultrasound) is another type of testing for use alongside mammography to help screen dense breasts. A number of leading manufacturers and research institutions have made significant advancements in mammography recently, particularly in the area of technology used in mammography. Also aiding in this objective are Hologic's Genius™ digital 3D mammography systems that use high-resolution 3D mammography to increase breast cancer detection rates by reducing false positivity (L. Wang, 2024) . A screening examination by mammography may have non-cancerous lesions interpreted as showing a cancer diagnosis, or may fail to identify a cancer that exists (missing diagnosis). Thus, radiologists generally miss breast cancer 10% – 30% of the time [2] [3] [4]. The proportion of positive nodule(s) which are subsequently found to be malignant after biopsy is an important determinant of the extent to which the screening examination under-identifies cancer (missed diagnosis). The miss rate for mammography is particularly high in dense breast tissue, at which point, the likelihood of having a breast cancer increases approximately 4–6 times when compared to the rate at which this same lesion might exist in a non-dense breast. Due to these factors, additional modalities, such as ultrasound and MRI, are utilized as adjunctive tests designed to enhance the sensitivity of mammography.

Ultrasound

Breast ultrasound (US) is inexpensive, easy-to-use, non-radiating, mobile and is usually very useful to determine if the breast mass is cystic or solid. There is well established diagnostic value of the US in palpable breast abnormalities particularly in heavy breast matter or concealed lesions. According to Dan et al., (2024) the US is supposed to be the preferred imaging modality to guide the biopsy procedure in breast.

The diagnostic yield of ultrasound is high for distinguishing breast cancer from benign lesions; it is also useful for preoperation assessment and is especially useful in detecting early cancer which has not been detected mammographically and/or clinically. Breast ultrasound becomes more beneficial as the breast gets denser and as younger women's sensitivity to mammography decreases. Breast Tissue Density is a common issue for most women, more than 50% of women under 50 have 50–75% heterogeneously dense or 75% of extremely dense glandular breast tissue. An estimated one in three women 50 and over also have dense breast tissue. However, only 30% to 48% of the mammograms of women with dense breast tissue will yield a normal result (9, 30). Moreover, interval cancer rates are significantly higher among women with dense breast tissue and women with dense breast tissue are 4–6 times more likely to be diagnosed with breast cancer than women classified as having normal ratios of glandular/fibrous tissue. Although breast ultrasound has not been used in any randomized screening studies, one cannot definitively say that using breast ultrasound will reduce breast cancer deaths due to the discovery of additional cancers. Therefore, even in such low-sized cancers, such as the nodal status, the range of tumor dimensions and the nodal state found in these ultrasound screening data is similar to that observed in the mammographic screening-detected invasive cancers.

Indications for breast ultrasound are as follows:

- Determining the characteristics of mammographic lesions as defined by the BI-RADS scale from 0-5
- Differentiating the types of palpable lesions between cystic versus solid, benign versus malignant

- Determining tumor characteristics: size, location, multifocality, and multicentricity
- Evaluating tumor response through longitudinal measurement to neoadjuvant chemotherapy treatment
- Providing guidance during an interventional procedure for a diagnosis Additional Screening for Dense Breasts on Mammograms ACR 3–4
- Young women under 40
- Pregnant and lactating women
- (Madjar, 2010) Further screening in high-risk patients.(Madjar, 2010)

MRI

Besides mammography and US, MRI breast is an indispensable modality. Given this, its primary uses include evaluation of reaction to pre-surgical drug therapy, ²¹screening in women with elevated risk for breast cancer, or staging of known cancer, and underscores its implications in this context. Unlike mammograms and ultrasounds, MRIs function in a different manner. This technique was introduced in the 1980s by both Hey wang et al and Kaiser and Zeitler. The rationale is that when neo angiogenesis occurs the blood vessels become leaky allowing contrast agent to be extravasated into the tissue faster, resulting in faster local enhancement. Because of the enhancement in breast imaging through MRI, the basic premise of this technique continues to serve as the foundation for all clinical protocols for utilizing an MRI. Therefore, to create images of diagnostic quality, it is necessary to use a special breast coil. The ⁶MRI is performed with the women lying in a prone position, allowing the breasts to hang loosely from the coil as they create the image of the breast. This allows the breast tissue to open up when capturing the image due to the motion generated by breathing, allowing for potentially more accurate determination of the existence of abnormal tissue without being influenced by the motion caused by breathing. It may be, but not necessarily, that four channels are needed in a breast coil; modern versions have 16 channels or more, and sometimes even more than the number of such channels in this special breast-coil for the axillary area. It must also be noted, however, that coils with more channels in general are able to yield a higher signal-to-noise ratio (SNR). In addition, the more

parallel imaging factors that are available will increase the speed of image acquisition.(Mann, Cho, & Moy, 2019)

Biopsy:

Liquid biopsy

Tissue biopsy has not provided complete information about gene alterations present in breast cancers using existing methods to make a cancer diagnosis. However, new methods involving liquid biopsy have enhanced the ability to provide an enhanced range of care for breast cancer patients with improved early detection and screening, prognosis, relapse detection, serial collection, and cost-effective long-term monitoring for both disease and treatment outcomes. An analysis of "liquid" biopsy samples can provide important information regarding many aspects of the tumor cell (CTC) (circulating tumor cell), circulating tumor (ctDNA), cell-free of RNA, Tumor Educated platelets, exosomes released from the tumor into the bloodstream. Therefore, there are multiple applications for each of these components, such as the sequencing of ctDNA for tumor genetic profiling and increased opportunities for personalized care and follow-up. It may also allow the early detection of minimal residual disease by measuring the amount of CTCs in plasma or ctDNA after the surgical removal of the curative tumor so that adjuvant therapy can be initiated before the recurrence of cancer occurs (Alimirzaie et al., 2019).

Vacuum assisted breast biopsy

Ultrasound technology is being used increasingly as part of the overall investigation process to aid in the evaluation of breast tumors and to assist with other diagnostic methods such as mammography and physical examination. Other established methods of evaluation include FNA (fine needle aspiration) and CNB (core needle biopsy) but newer methods of Biopsy are VABB (vacuum-assisted breast biopsy) has many advantages and can be performed under guidance.

Sentinel lymph node biopsy

Lymphatic mapping and sentinel lymph node (SLN) biopsy methods are newly developed procedures that provide accurate critical staging material to the patient while reducing the potential morbidity of traditional complete axillary dissection, and warrant further study. As any new surgical procedure, the SLN mapping technology for breast carcinoma should prove to be efficient, reduce risk, and lower the costs and resources associated with the procedure. In the mapping literature using dye, the two agents of most relevance to the field and mentioned are lymphazurin and patent blue dye. Biochemical stability would be the argument for lymphazurin, specifically Lymphazurin 1%, which has been shown to have no pharmacological action and is a first in class, preservative-free contrast agent that can be used for identifying lymphatic vessels (Bass, Cox, Ku, Berman, & Reintgen, 1999).

Histopathology

In tumour diagnosis, histopathology is a key important part that helps the pathologists accurately categorize the section of tissue as benign or malignant and provide additional information like tumour aggression and grade. Since then, immunohistochemical (IHC) staining to assess contourstone biomarkers, like oestrogen binding and progesterone receptor (PR) state, remains in place to steer therapy for the patient. Current guidance on the assessment of biomarkers is relatively restricted because there are various types of tissue fragments, and the assessment of the clarity of the biological indicators differ from one alternative to another. Inconsistencies might arise between diverse studies, as suggested in the conclusion, thus making the search for “modern” biomarkers and their use for diagnostics and therapeutic interventions more interesting. Of great importance is the suggestion that the initial clerical and logistical limitations facing digital tissue analysis and computer-based digital analysis in clinical medicine have started to be overcome. With greater accessibility of improved computerized visuals, potentially, more precise analysis of morphological and biological diversification can be performed with less inter-professional variation amongst observers and greater reproducibility. The potential of morphological regularity evaluation has been demonstrated; also in the field of biological evaluation, AI application provides results more accurately and, based on the results of the histopathological examination of slides, more standardised and reproducible than would be obtained by purely manual examination of the slides.

Although orthodox histology still ³³plays a major role in detection, the incorporation of machine learning-based methods is consistently enhancing accuracy and consistency in the histo-pathological assessment, showing promising advances.

Digital pathology

Molecular testing for breast cancer is rapidly evolving, and now forms a more significant aspect of breast cancer treatment than before. Molecular assays and testing methodologies designed to help identify ⁶the diagnosis, prognosis and treatment of breast cancer are now ¹⁰⁶an essential component of precision medicine-related diagnosis, prognosis and treatment options for breast cancer. There are also the management and collection of single handwritten slides. Some pathologists have reported that they are capable of viewing WSI images using custom software at the clinical magnification level. [12, 24] ²To facilitate the safe and efficient use of digital protocols for diagnosis, a clinical validation evaluation of the diagnosis of breast carcinoma made from a digital slide was performed in compliance with rules established by the College of Pathologists. Three pathologists were trained in digital pathology, to start with 20 difficult test cases. The level of clinical agreement found between glass slides and digital media was 98%. Unfortunately, only 8% of the total number of confirmed cases of identical relevance was clinically significant ($p < .05$). Furthermore, additional training can provide a pathologist with confidence to use digital imaging tools.

²Artificial Intelligence in Breast Cancer Diagnosis

The emergence of easily and inexpensively available technologies to have the ability to analyze digital evidence and/or store digital information, in combination with expanding capabilities of inexpensive computers' growing computational power and the growth of available cloud resources to store digital evidence, will continue to build exponentially. Further, artificial intelligence (AI) is continuing to evolve and be integrated into many common gadgets including computers, mobile devices, cars, and smart watches for a large number of functions including robotics, spam filtering, speech recognition, etc. However, there are specific types of AI: AI machines can be created in a way that allows them to replicate some of the functions of humanity.

Deep learning

Deep Learning (DL) algorithms are relatively new advancements of computer aided detection (CAD) or diagnosis (CAD) methods derived using neural network (NN)

architecture(s) that are analogous to form an NN processing model of cognitive function of the human brain. DD decision-making from the application of DL will be increasingly based on the information from different sources. on data.

Histopathology image analysis

Cruz-Roa and colleagues have likely developed a powerful method for histopathological image analyses based on a method called Histopathological Image Analysis (HASHI) with a Convolutional Neural Network (CNN) based technique that uses probabilistic gradient-based sampling and quasi-Monte Carlo methods to identify whether an image of whole slide invasive ductal carcinoma exists or not. (Ivanov et al., 2025)

Genetic and Molecular Testing

As of 2009, there are several genetic tests commercially available on the market that evaluate the presence of genes associated with cancer predisposition. Genetic tests can be performed using either a blood sample (liquid biopsy) or a tissue biopsy (saliva or cheek swab) to identify cancer-related mutations. Before the advent of next-generation sequencing (NGS) technology, these types of tests were performed one at a time.

There are multi-gene panels that may be carried out in patients diagnosed with breast cancer such as panels for the purposes of providing a driver for diagnosis of invasive cancer (for example, Oncotype DX; MammaPrint; Prosigna; Breast Cancer Index; etc.), ductal carcinoma in situ (for example, Oncotype DX DICS; DCISionRT; etc.), and PIK3CA mutation and NTRA fusion as companion diagnostics. Such panels will assist in determining eligibility for immunotherapy, that is, immune checkpoint inhibitors (anti-PD-L1 monotonal antibody SP142 and 22C3) and finding MSI, dMMR, and TMB. Molecular testing is one of the most advanced methods for diagnosing and assessing the prognosis for various forms of cancer, and most of these techniques depend on the type of cancer. Among all solid tumors, breast cancer was among the early cancers to apply molecular biomarkers for pathologic evaluation in addition to morphological assessment. This is why there is a very long track record of using HB and HER2 or EGFR overexpression and/or amplification for prognosis and treatment decision-making for breast cancer.

The oncotype DX

The RS is determined from 16 cancer-associated and 16 reversing gene control or reference values used in the RS. The RS predicts a female patient's likelihood for distant recurrence, if she has stage I ER + / HER2 - invasive breast cancer. The RS has been

developed by the Oncotype DX laboratory located in Redwood City California through their enhanced services division. Each of the 16 genes provides a score based on their weighted contribution to the total score. Thus, the larger the RS, the more likely the possibility of distant relapse and, at the same time, the opportunities to improve the condition of the patient through hormone-free chemotherapy will be more numerous.

Oncotype's RS Development began with advanced (positive-node) BC. A low (RS score < 18), intermediate (RS score > 18 to 30) or high (RS score > 30) risk was developed at a subset of patients as an independent predictor of metastasis without regard to the patient's tumour stage or age. The foundation of using RS rests on an NSABP-B20 retrospective analysis which revealed that there is a statistical difference between chemotherapy administered to the RS greater than 20 patients group and a number of metastases in patients with RS scores of 21-29 than those in patients with RS scores 29 or lower. That means that no influence of the RS was observed on distant recurrence of breast cancer for patients having RS score of less than 20. To determine the predictive efficiency of the RS for the treatment of patients with intermediate RSs and solve an ongoing controversy on patients with intermediate RSs, the TAILORX Study: Phase III clinical trial was conducted. A number of retrospective studies to prove the diagnostic efficacy of RS for the treatment of HR+ node-negative IBC patients (analysis of a subgroup of patients according to randomisation and administration of adjuvant chemotherapy and endocrine therapy) resulted in contradicting conclusions. The retrospective studies used a variety of patients who had either N1 alone or both N1 and N2 staging with their adjuvant chemotherapy as one of the treatment subgroups. Out of the entirety of the retrospective analyses, the secondary analysis yielded a relationship between the RS and the relative risk ratios for the endpoints of DFS and OS, not only for N1, but also for those N2 treatment subgroups.

Multigene assay

There are numerous, proven, commercially available MGAs for the HR+ patients with early-stage IBC and DCIS, see Table 1. It is essential to mention that these tests use many different genes and return a score which provides an estimated prognosis (recurrence/survival) and prognosis (predicting likely response to treatment).

Mammaprint and Blueprint

The genetic activity profile for Blueprint assay includes 80 genes (Agenida - Netherlands), allowing breast cancer tumours to be described as being part of one of three intrinsic breast cancer types- basal-like, luminal A or luminal B. The MammaPrint and Blueprint assays can be obtained upon request through the clinician in the US from

Agendia. The combined report contains the MammaPrint assay's binary risk of recurrence (low or high) as well as MammaPrint, results of the BluePrint assay for molecular subtype.

BCI

The European Directive 93/42/EEC classifies ¹⁰ MammaPrint BluePrint Breast Cancer Recurrence and Molecular Subtyping Kit as CE marked in accordance with European Law. Therefore, this product may be sold and used throughout all countries that comprise ¹⁰ the European Economic Area (EEA). However, ¹⁰ it is not currently available for sale as a kit with an FDA approval in the US. (Sun, Wu, Bean, Hagemann, & Lin, 2021)

EndoPredict

In addition to the determination of the ³ clinical and pathological dataset (tumor size and lymph node metastasis), ¹ the EndoPredict test evaluates the expression levels of ¹² 12 genes whose combination provides the generation of the EPclin score. The EPclin score has been confirmed to predict the chance of recurrence as well as ¹²⁵ identify the type of the adjuvant therapy in use among patients classified into the defined ¹²⁵ high-risk and low-risk groups of ER+/HER2- breast cancer (76). The previously mentioned ABCSG-6/8 study that involved postmenopausal patients with ER+/HER2- tumours treated with endocrine therapy only also evaluated the EPclin score and distant recurrence-free rate (DRFR); the patients within the minimal EPclin score groups had a significantly better DRFR than those in the high EPclin score groups regardless of lymph node status with statistically significant differences for both strata, demonstrating the prognostic utility of the EndoPredict assay.

Genetic Testing for Metastatic Breast Cancer

As per published literature, about 3-8% (depending on study design) of ³² breast cancer patients will initially be diagnosed with metastasis. Breast cancer with metastasis presents with a ³² very poor prognosis since the 5-year survival rate exceeds 75% in such patients (76). However, despite the difficulty in curing these patients, it is possible to improve treatment outcomes with new approaches ⁹⁷ aimed at improving the patient's quality of life and extending her lifespan.

Genetic Testing for Metastatic HR-Positive Breast Cancer ⁷

Many patients diagnosed with hormone receptor positive (HR+) metastatic breast cancer have either a PI3KCA mutation or an ESR1 (mutated estrogen receptor) mutation. when examining patients who are resistant to aromatase inhibitors,

approximately 33% have evidence of ESR1 mutations (D538G, Y537S and Y537N) [90,91]. The evidence that suggests ESR1 mutations as a primary mechanism of aromatase regulatory drug lack of sensitivity had been obtained from clinical trial investigations comparing the efficacy of fulvestrant⁵⁴ and the aromatase inhibitor exemestane in HR+ metastatic breast cancer patients who have ESR1 mutations identified via circulating tumor DNA (ctDNA) sampling; the EFeC and SoFEA clinical trial studies support that fulvestrant is superior to exemestane. Studies have documented that when the PI3K pathway is activated, this leads to uncontrolled cell division or proliferation and has nothing to do with the stimulation of estrogen [95]. There is a possibility that by suppressing estrogen production in this large group of patients, the PI3K pathway will then be activated, and PI3K has now become an anti-cancer target.¹⁷

The efficacy of the PI3K inhibitor buparlisib, used in combination with fulvestrant, for patients with metastatic breast cancer and a PIK3CA mutation was reported in the BELLE-3 trial. Additionally, based on preliminary evidence from SOLAR-1, the PI3K inhibitor alpelisib may also be effective for individuals having advanced breast cancer.³ These data provide strong rationale for the accelerated approval of alpelisib as treatment for individuals with HR+/PIK3CA-mutated metastatic breast cancer by illustrating the significant prolongation of tumor-free patient outcomes within the Alpelisib cohort. Examples of mutations related to invasive breast cancer presented by next generation sequencing (NGS) include H1047R, E542K and E545K (99). Tissue from a metastatic site, and/or ctDNA from blood, may be utilized for the evaluation of PIK3CA mutational status utilizing NGS technology.

³⁹ Genetic Testing for Metastatic HER-2-Constructive Breast Cancer

The use of monoclonal antibodies such as trastuzumab and pertuzumab on HER2+ breast cancer patients has been found to be effective. However, antibody-drug conjugate (ADC) therapy comprises a combination of monoclonal antibodies and chemotherapy drugs. Some studies show that ADCs such as T-DM1 and T-DXd are equally effective in treating HER2+ metastatic breast cancer (MBC) with low HER2 levels.⁷

¹ Genetic Testing for Metastatic Triple-Negative Breast Cancer (TNBC)

Furthermore, there is some correlation between breast cancer development and mutations of BRCA1 and BRCA2 genes. Additionally, it could be argued that BRCA gene mutation might cause the formation of early breast cancer despite the BRCA mutation being mostly seen among triple-negative breast cancer metastatic patients. It¹¹⁷

has been established that there is involvement of the BRCA genes in BRC1 and BRC2, as they both play a role in repair of DNA double-stranded breakage by homologous recombination and cell cycle regulation. When mutation occurs in BRCA1 or BRCA2 genes, the tumor suppressor quality is diminished, which results in increased rate of cancer development. Currently there are >20,000 mutations in the BRCA1 and BRCA2 genes, with mutations located randomly throughout the genes. Thus to obtain complete analysis of BRCA1/2, it is critical that the entire coding sequence of BRCA1/2 be sequenced to ensure all areas of the genes outside of the two known hot spots are included [132,133]. Results from the OlympiAD and EMBRACA studies indicate that PARP inhibitors may help prevent cancer recurrence in patients with HER2-advanced breast cancer who also have BRCA mutations. Compared with patients receiving chemotherapy, those treated with olaparib demonstrated an improved median progression-free survival (3 months) and were able to enter remission at double the rate of patients receiving chemotherapy. Talazoparib also demonstrated an improved 3-month median progression-free survival in patients with germline BRCA mutations. To assess clinical benefit for patients diagnosed with MET BC and negative HER-2, psychological assessment of patients' BRCA1/2 gene mutations is recommended (Yang, Li, Huang, & Zhou, 2023).

Biomarkers

Biomarkers are characterized by either being tissues (for instance, prostate carcinoma specific genes or with evidence for EGFR kytase mutations in non-small cell lung carcinoma). Thus Several scientists have provided evidence to support their definition of tissue biomarkers via these examples of these types of biomarkers. To illustrate, the NCI characterizes a biomarker as being detectable within the bloodstream and other bodily fluids; it's presence or absence reveals whether or not normal cellular activities are taking place.

Biomarkers capable of determining the likelihood of the optimum treatment and subsequently monitoring the success of that treatment during the initial period would be highlybeneficial.

ER is one of 6 nuclear hormone receptors belonging to what is referred to as a steroid receptor and acts as a transcriptional regulator when the hormone E2 is available.

MUC1 serves as a biomarker due to its overexpression in >90% of ER+ breast cancers. In addition, existing research indicates that MUC1 may be a viable candidate for breast cancer immunotherapy. However, ⁸³it is important to keep in mind that pre-clinical trials have demonstrated that MUC1 rBCG (Bacillus Calmette-Guerin) vaccines have elicited an anti-MUC1 immune response.

As a result of this the approaches described above could potentially impact PIN1 activity associated with HR in BC, and subsequently alter the stability of PIN1. There is a direct relationship between hormonal therapy resistance and GSK3-mediated phosphorylated sites on HR (hormone receptor). Recently, GSK3 inhibitors such as CG0009 ²have been shown to reduce proliferation and overall survival ⁴²of various breast cancer cell lines.

Mutations of the ESR1 gene (a generic/every breast cancer-associated mutation of the ESR1 gene) that are associated with producing the HR protein have primarily been demonstrated to be a mechanism of endocrine (hormonal) resistance.

HER2

The HER-2 gene that encodes a class of growth factor receptors (type 1) is located at chromosome 17, specifically q21. HER-2 also occurs ³with Hormone Receptor (HR), another well-known breast cancer (BC) biomarker. The majority of the growth factor ³receptor family have similar receptors that participate in the inter-cellular communication between cells through the modulation of gene transcription – usually through the phosphorylation of various membrane proteins and/or protein kinases (non-duplex proteins) found in the cytoplasm of each cell. This is supported by the conclusion that HR and HER-2 are absent from cancer cells, and expressed in epithelial cells of various tissues, including but not limited to, the lungs, bladders, pancreas, and breast and prostate. Therefore, HR is typically over-expressed (greater than twenty to thirty percent) in primary tumors.

PR

PR is a PGR gene of the progesterone receptor located on the long arm of chromosome 11 (11q22-23), composed of 8 exons and 7 introns, which has demonstrated the greatest

benefit of a breast cancer tumor marker that correlates with the ability to determine the effect of hormonal care.

CEA and CA

When considering breast cancer, patients may be evaluated based on the following serum tumor markers for their individual risk assessment: CEA, CA-15-3 and CA125. The levels of these tumor markers, when sampled preoperatively, can also be correlated to important prognostic data pertaining to the patient's tumor size, axillary lymph node status and hormonal binding site state.

CTCs

Rare cancer cells, known as CTCs, as they are found in the peripheral blood and are classified as an active role in the progression of tumors. Several studies have shown that CTCs are an major predictive biomarker in breast cancer [14]. Therefore, while the use of CTCs cannot be taken to confirm a patient's prognosis with certainty, increasing numbers of CTCs means that a patient has decreased odds of overall survivability; persistence of these cells during treatment can also be indicative of poor outcome of medication rousing breast cancer.

Ki67

The measure of the Ki67 tumor can also be used to indicate the mitotic activity of the tumor, or that the tumor is "growing," and that the Ki67 has a short half-life. Research indicates that Ki67 is an excellent and reliable marker for cellular proliferation, as well as a predictor and prognostic factor for breast cancer (BC) recurrence and BC prognosis. Additionally, Ki67 is one of the 21 gene markers used in Oncotype DXTM for providing information about BC recurrence and the potential of chemotherapeutic intervention efficacy in the post-menopausal ER+/node-negative breast cancer patients.

Cyclin D1

Cyclin D1 is one of the cyclins belonging to the D-type family and is an important regulator of the cell cycle. Cyclin D1 does this by functionally activating cyclin-dependent kinases (CDK) in order to control all cellular events between G1 and S phase (Bishop et al, 2021). More specifically, when cyclin D1 binds to CDK4 and CDK6, the result is the hyperphosphorylation of retinoblastoma (Rb) protein.

pD-L1 and TILs

PD-L1 (CD274) is an immunologically active transmembrane glycoprotein of particular interest in many normal host cells that act as immunological signaling molecules. PD-L1 has the ability to interact with PD-1 which acts to modulate or suppress the host's immune system response to stimuli.

Surgery

Women detected with breast cancer who opted for either lumpectomy with adjuvant radiation therapy or lumpectomy alone as the preferred treatment were found to have 14% and 39% cumulative recurrences of breast cancer (0.05% ⁵⁷ $P=0.51$) and (0% $P=0.74$) respectively. In addition, there was very little difference between women receiving total mastectomy and those receiving lumpectomy plus adjuvant radiation therapy for developing new cancers in other parts of their body (1.0% ⁴⁰ $P=0.51$; 95% CI, 0.9 to 1.2) and (0% $P=0.74$; 95% CI, 0.8 to 1.1). Additionally, their respective rates were expected to be similar (1.0% for women with lumpectomy only and 0.97% for women with total mastectomy; $P<0.001$). Overall, based upon these data many healthcare providers believe lumpectomy plus radiation ¹⁴ therapy is an effective treatment option for women diagnosed with breast cancer.

⁴ Axillary lymph node dissection (ALND) has ceased to be the most common diagnostic tool for assessing the disease extent ⁷ in breast cancer patients since the approach for determining the extent of the disease has been altered. The preferred diagnostic tool in contemporary oncology is sentinel lymph node dissection (SLND) due to its minimal invasiveness (Giuliano et al., 2017). The proportion of women who had an ALND performed decreased from 94% in 1998 to 36% in 2004 and among those patients who had sentinel node metastases, 68% had an ALND performed in 2004.⁸ Although the ALND is a valid and useful technique for controlling the cancer within the axillary region, it can produce numerous side effects, including lymphedema, numbness in the arm, axillary web syndrome, and limitations on upper extremity mobility. The question arises whether any patients with positive sentinel nodes should have undergone a completion ALND based on how they presented with breast cancer and what form of systemic therapy was used based on the biology of their tumour. Recent advances in the area of oncoplasty and reconstruction now allow for breast reconstruction to take place soon after a patient undergoes a mastectomy either via the use of an implant or via reconstruction using body tissue from a different area (Rao, Euhus, Mayo, & Balch, 2013).

Chemotherapy

When given ¹⁹ in addition to the primary treatment of breast cancer, adjuvant treatments, which may include endocrine therapies, chemotherapies, and anti-HER2 therapies, reduce the likelihood of recurrence. The availability of adjuvant systemic treatments, as well as their significant and dramatic contributions to decreased mortality rates throughout the world due to breast cancer, are documented. Finally, early investigations of trastuzumab (HER-2/neu humanised monoclonal antibody) have demonstrated ⁵⁸ its application in the management of breast cancer. Trastuzumab has been referred to as the first rationally developed targeted biologic that has shown efficacy in randomized clinical trials. Therefore, it can be asserted that ²⁶ trastuzumab is the standard therapeutic agent for treating breast cancer patients.

The use of ⁶⁵ neoadjuvant chemotherapy (NAC), which is now considered the standard for treatment for metastatic breast cancer (MBC), improves outcomes for most MBC patients— particularly in TNBC and HER2 overexpressers – by aiding the surgeon in determining optimal surgical technique and procedure through pre-operative prognostic data. There have been some studies to show ¹²¹ that patients who have achieved pCR (pathological complete response where there is no ⁷⁸ invasive breast cancer found in the breast or lymph nodes following the use of neoadjuvant chemotherapy) exhibit higher EFS and OS (Event-free survival and Overall Survival) ¹²⁶ (Anampa; Makower; Sparano; 2015). In addition to anthracycline drugs and nonanthracyclines, other agents have also shown to be highly effective in first-line treatment of biopsy-proven MBC one year following treatment (e.g., with anthracycline) as well as previous anthracycline exposure with an adjuvant therapy and subsequent development of MBC. As such, there is a reported response rate for MBC using both types of drugs between 30% and 40%. The evidence supports that both types of chemotherapy play a role in the adjuvant setting; however, the potential for significant toxicity also exists. Meta-analyses that have evaluated thirty studies of PEGylated liposomal doxorubicin suggest that the most likely reasons for achieving a therapeutic effect would be its pharmacokinetics and its ability to target the breast tumour. The selectivity for breast tumours is thought to be due to the enhancement of the permeability and retention-effect of the breast microenvironment, which is likely due to the increased permeability of the tumour blood capillaries and impaired lymphatic drainage (Waks & Winer, 2019).

Radiotherapy

According to the evidence, including data from a large series of random clinical trials as well as objectives similar to those reviewed above, radiotherapy to the breast after breast preserving surgery halves the rate of breast tumor occurrence and halves the threat

of breast cancer death. Therefore, it is likely that these proportional benefits are fairly consistent among a variety of groups of women (Hennequin et al., 2022). Radiotherapy has an absolute benefit that varies per patient. The extent of the absolute benefit will be dependent on certain characteristics of the patient at the point patients are managed and are able to reach the stage therapy choices must be made. Internationally, the combination of two larger outcomes; women receiving RT had their risk of any first time recurrence (locoregionally or distantly) decreased from 35.0% to 19.3% over ten years, their risk of dying because of BCY had – over fifteen year total – decreased from 25.2% to 21.4%. The risk of a woman having their first time recurrence, as well as developing a fatal case of BC, with pN0 (n=7287) had their first time recurrence risk decreased from 31.0% to 15.6% and their risk of dying from BC decreased from 20.5% to 17.2% (absolute risk in mortality; $3.3\% \pm 0.8\%$ to 5.8% ((being the broad range of risk)), (p=0.005)) (Group, 2011).

For patients under 50 years old, an additional round¹⁶ of radiation therapy to the site of the tumor bed will be given after surgery (boost). Based on available evidence and clinical judgment, a physician may make a routine recommendation for selective -breast radiation as an option for patients diagnosed with ductal carcinoma, and will discuss with an individual patient that particular option based on the unique characteristics of their situation. There should be a systematic approach to the use of adjuvant (post-operative) radiation¹⁰³ after breast protecting surgery in patients diagnosed with ductal carcinoma. Following a patient's mastectomy, if they present with pT3-T4 tumor stage then they will need to receive chest wall radiation therapy. In addition, if there is axillary lymph node involvement, the patient will need to receive chest wall radiation irrespective⁷ of the number of lymph nodes involved. After a patient has had neoadjuvant chemotherapy and mastectomy, chest wall radiation is appropriate in any of the following circumstances: pN0 disease, clinically T3-T4 disease prior to chemotherapy, radiographically T3-T4 disease prior to chemotherapy or node positive disease prior to chemotherapy (Haviland et al., 2013).

Immunotherapy

within the specified immunotherapy drugs used to treat breast tumor will be monoclonal antibodies that target HER2, PD-1, and PD-L1; the latter two types of drugs are referred

to as immune checkpoint inhibitors (ICIs). By inhibiting the binding of the checkpoint proteins (and, thus, the PD-L1/PD-1 receptor) and their partners (T-cell receptors, for example), the ICI allows the T-cells to kill the cancer cells by preventing the “turning off” signal from being sent to them (Schmid et al., 2020). Cancer cells are ready to evade the defensive system through their ability to change and avoid detection, which makes them immune resistant to destruction by the immune system. One way this occurs is through the promotion of PD-L1 binding to PD-1 in order to block access of immune cells to the cancer and kill it. When an ⁵⁰immune checkpoint inhibitor (ICI) is ⁵⁹used to block the linked of PD-L1 and PD-1, immune cells can then recognize and destroy cancer cells (Schmid et al., 2020). Thus, ICIs remove the “brakes” of the immune system's anti-tumor response toward cancer. As a result, the defensive process will now be able to recognize and kill the cancer cells after having either previously not been able to identify them or, having therefore been able to live together in harmony with the immune system before the administration of ICI drugs. This mechanism of ICI action, in addition to providing a therapeutic benefit from the deletion of cancerous cells, also provides an explanation for the immune-related adverse effect associated with the use of the checkpoint inhibitors, and therefore, for the occurrence of autoimmunity, which can be severe, permanent, and in rare cases fatal to the patient (Dvir, Giordano, & Decker).

According to data obtained from the IMpassion130 trial, patients receiving atezolizumab plus nab-paclitaxel and possessing PD-L1 staining $\geq 1\%$ on the VENTANA SP142 IHC Assay displayed a PFS increase of 2.5 months. This information may have served as an impetus for FDA to grant the accelerated approval of Atezolizumab in March 2019; however, recent evidence seems to indicate that patients with TNBC ³¹who are at high risk for recurrence (stage II and III), will be treated using pembro (KEYNOTE 522), using a regimen of Neoadjuvant Paclitaxel-Carboplatin + Pembro followed by Doxorubicin-Cyclophosphamide + Pembro. It could also be argued that the KEYNOTE 522 trial demonstrated a greater rate of pCR with the use of pembro (65%) than without pembro (51%).

Targeted therapy

The molecular and biological underpinnings of breast carcinoma were explored as new therapies are being developed to target these pathways. Patients suffering from breast cancer can consider targeted therapies such as targeted hormonal binding site regulators. Therapies available to treat breast cancer include anastrozole, letrozole and exemestane (aromatase inhibitors), tamoxifen, raloxifene, and lasofoxifene. They have all been studied through multiple clinical trials, which included both treating as well as preventing breast cancer. Clinical trials so far have suggested that ⁸⁵ tamoxifen and raloxifene lower ⁶⁶ women's risk of developing breast cancer. Some researchers believe that AIs may prevent (ER+) breast cancers more effectively than either tamoxifen or raloxifene, based on early findings from studies evaluating estrogen-blocking agents for therapy of breast cancer (Higgins & Baselga, 2011).

Recent reports indicate that new methods of targeting ¹³ human epidermal growth factor receptor 2 (HER2) are fighting ⁶⁶ tumors that continued to grow despite previous treatments. Trastuzumab-DM1 (T-DM1), a type of monoclonal ² antibody-drug conjugate (ADC) with selective binding to HER2-positive tumors, is one such example. T-DM1 (or trastuzumab-DM1) is composed of trastuzumab, an anti-HER2 monoclonal ¹² antibody, covalently linked to maytansine (also known as DM1), a cytotoxic antimicrotubule agent. Because DM1 has the capability to kill cancer cells, powerful cytotoxic agents can be delivered directly into HER2-expressing ¹² tumor cells (Den Hollander et al, 2013).

There ⁵ are a number of potential drugs currently under evaluation for use as chemopreventive agents for TNBC. The majority of them have been the subject of clinical studies for the purpose of administering these chemopreventive agents, however, the actual data is limited. Of all the drugs that have been studied and considered potential chemoprevention options for ⁵ triple negative breast cancer (TNBC), poly(ADP-ribose) polymerase inhibitors are among the most frequently investigated, and are thought to be strong candidates for use as chemoprevention options. Other options available as chemoprevention include both vitamin D and rexinoids, as they are thought to improve the efficacy of other cancer prevention options. In addition, there are also several retinoid compounds that have been developed that are being studied as possible chemopreventive agents for TNBC. Retinoids are naturally occurring compounds derived from vitamin A, which bind to retinoic acid receptors (RARs) and regulate the transcriptional output of genomic components play

a role in development (in part), differentiation, and homeostasis (Alvarez, Valero, Horthobagyi, 2010).

Personalized medicine

Consistently, the data indicates that patients respond differently to anticancer agents and that harnessing new genomic, transcriptomic, and proteomic technologies to enhance the insight of the molecular basis of cancers can facilitate for identification of specific genes associated with individual variations in clinical outcomes and/or response to drugs. Thus, there is strong evidence to suggest that personalized medicine has transformed the health care paradigm by facilitating the use of an individual's genetic information to optimize the efficacy of drug therapies and thereby reducing the number of adverse drug events (ADEs).

Management strategies for the most prominent class of breast disease, luminal A, include primarily the use of anti-endocrine therapies - since 40%-60% of luminal A cancers have an **estrogen receptor** positive (ER+) and/or **progesterone receptor** positive (PR+) phenotype and an amplification of the HER2 gene. This population also shows low pathological grade (grade 1-2) and low proliferation rates, and therefore has shown low response rates to chemotherapy but is responsive to endocrine therapy (i.e. anti-estrogen therapy) (Cho, Jeon, & Kim, 2012). Luminal B, which includes HER2 positive (n=15% of breast cancers), also shows higher grade and proliferation than luminal A, so therefore has higher response to chemotherapy than anti-estrogen therapy. Tumors considered HER2-overexpressing (n=10% of breast cancers) also have a higher pathological grade than both luminal A and B, and therefore have poor overall prognosis until the emergence of targeted therapy was developed for HER2; although all of these subtypes (luminal A, luminal B, HER2 overexpressing) are chemosensitive, but vary in terms of aggressive nature and prognosis following chemotherapy. Basal-like cancer is a subtype of breast cancer that includes 10-25% of cases; the basal-like having a profile of ER- /PR- /HER2- but having to include 15-45% of tumors in that population of ER+/HER2+/-; additional information in favor of the claim of existence of basal-like breast cancer is provided. Of note, the data indicate that basal-like breast cancers may also be chemosensitive, and they are strongly associated with highly aggressive behavior and poor prognosis being met by 3-10% of patients for classification as normal-like (Jeibouei et al., 2019).

Challenges in Breast Cancer Management

Drug resistance

This finding is based on the principle that continued drug-resistance remains the primary limit to finding cures for patients with cancer. Drug-immunity in cancer is very similar to drug-resistance in infectious diseases. In both areas of medicine, there is a problem of rapidly replicating intrinsic and/or extrinsic factors that create drug-resistance. Initially, there was tremendous enthusiasm for anticancer chemotherapy once early chemotherapeutic agents produced rapid responses in patients, but this enthusiasm quickly waned when patient tumors developed resistance and subsequently relapsed after achieving complete remission. As a result, the first response to this challenge of drug-resistance associated with the use of single-agent chemotherapy became to combine multiple agents with non-overlapping mechanisms of action (i.e., polychemotherapy) (Higgins & Baselga, 2011).

With varying degrees, some patients experience the development of secondary tolerance against their treatments due to the progression of their disease as well. Some research has shown that there are mechanisms through which tumors will stop responding to a treatment after originally being sensitive to it that have been documented in HER2-positive tumors (Vasan et al. 2019). Evidence suggests that there are three potential mechanisms that may lead to the re-emergence of tumors or the development of secondary resistance: loss of expression of the therapeutic target following long-time exposures to the target, downstream mutations that result from the target being activated, and alternative mechanisms that stimulate cellular replication. Primary resistance can occur because the cell was never established to depend on a therapeutic target (Lange & Ghosh, 2020). More recent data from the literature indicate that activation of compensatory pathways in a cell can by-pass the effects associated with blocking a single target/pathway on that cell. An illustration of this occurred when the PI3K/AKT/mTOR pathway was blocked causing multiple survival pathways to be activated and provide compensatory survival functions (e.g., IGF1R, HER2) (Kinnel et al. 2023).

Late diagnosis

The key to overall health, especially regarding the development of disease, is early diagnosis, including breast cancer. In general, when comparing patients detected at an early stage of cancer to those diagnosed at a later stage, the patients diagnosed earlier will have fewer complications and an increased ability to live longer. Access to

adequate health care, specifically primary care and mammography, is a significant element of early detected of breast tumor since there are various degrees of availability to the health care system due to financial, socio-cultural, or geographical influences; however, we do not currently know the relative impact of these on ³the likelihood of having late-stage breast cancer (Hisham & Yip, 2004).

There is evidence that groups disadvantaged by income and race ¹⁵are more likely to experience late-stage breast cancer diagnoses based upon previous research. Women with a lower educational attainment have a higher probability of being diagnosed. While some have argued that African American and Latina women are at greater risk of being diagnosed late in their disease process than other races or ethnicities, these disparities are primarily attributed to differences in social and economic status and access to health care. Evidence reveals that low-income, ethnic/racial minority women often present later than other female with breast tumor, due in part to their firm belief in Western medicine, an emotionally negative understanding of breast tumor a shortage of awareness based on poverty and lack of education, and fear and denial. The outcome of this research implies the importance of carrying out longitudinal research in communities in order to better study the demographic data and characteristics of patients suffering from breast cancer in the community in an attempt to understand why such cases are delayed (F. Wang, McLafferty, Escamilla & Luo, 2008).

Treatment side effects

Adjuvant therapy refers to all types of therapies used to treat patients who may develop recurrent cancer/relapsed cancer following surgery (removal of tumor) or following completion of chemotherapy or radiation (all others). The benefit of adjuvant therapy to prevent recurrence following surgery is well established for several types of cancers; Paxil and other adjuvant agents have demonstrated efficacy as adjuvant therapy in the care of recurrent hormone-receptive (HR+) breast cancer (Ward et al 2018). Other types of cancers also demonstrate gain from additional intervention. However, studies have indicated that approximately 30% of women have to discontinue adjuvant therapy prematurely due to adverse effect. When compared to other treatment modalities and/or regimen, many side effects from agent to agent will often differ; Nevertheless, with some patients, the severity of side effects was managed to the point where quality of life did not beget early discontinuance vs an alternative agent (Partridge et al 2001)

Furthermore, it's plausible that hot flashes (vasomotor) will continue to be the most prevalent side effect from all of the adjuvant endocrine trials. As such, it is assumed that certain side effects that are common (i.e., vaginal discharge, vaginal dryness,

dyspareunia, and joint pain) will have a variable incidence in comparison to the AIs with tamoxifen use. Hot flashes, joint stiffness or pain, decreased bone density, and sexual dysfunction such as vaginal dryness are among the side effects that may develop from the use of aromatase inhibitors; however, the absence of the anthracycline-based drug in a treatment plan equates with fewer short-term toxicities of grade 3 to 4 compared to an anthracycline-based treatment plan (Franzoi et al., 2021).

Limited Healthcare Access in Developing Countries

However, the limited resources of developing countries also reflect an absence of one common strategy for diagnosing breast cancer and, generally speaking, most of the population uses a public health system for its cancer diagnosis,(Unger-Saldaña, 2014). It is hard to compare the cancer incidence and deaths between developed nations and underdeveloped nations because of the vast difference in healthcare systems. 60% of breast cancer deaths are from underdeveloped nations, whereas the United States has approximately 249260 new diagnosed ¹¹¹cases of breast cancer each year, and the breast cancer ¹¹¹mortality rate is decreasing over time.

As such, ¹⁰²the attempt has been made to objectively identify the variation in complexity of the breast cancer health care system. The Breast Health Global Initiative (BHGI) has been developed to create a system of metrics by which to quantify the quality and organization of breast cancer health systems.

At the lowest rate of care, patients are encouraged to do breast self-examinations, while outpatient mammogram and ultrasound are able to provide diagnostic images of breast cancer. Patients are able to access ¹²⁸diagnostic mammography (diagnosing). As well this will give them the opportunity ¹²⁸to screen for breast cancer correctly and at the best level of support they can receive. Breast cancer will have a good prognosis based upon the research that has been published to date. Breast cancer will have a survival rate of 73 percent in developed nations and a survival ²⁰rate of 57 percent in developing nations with a 5 years survival rate of 89.7 percent ²⁰in the United States. The incidents of breast cancer will be lower in developing countries, but that the vice versa for breast cancer. Because there are logistical and macroeconomic challenges blocking initial detection of breast cancer in developing nations (Panieri, 2012).

Recent advances and Emerging Trends

AI in diagnosis

Many researches have focused on creating AI algorithms that detect and classify breast cancer. ² Cruz-Roa et al. developed a theory based on convolutional neural networks (CNN) that can classify images (patches) of breast cancers based on slides of woman (WSI) to determine how far through breast tissue, the cancer has spread; this relies on an AI classifier. Han et al. reported that the average accuracy of a deep learning algorithm was 93.2% against the other eight subtypes (four benign and four malignant) listed on their dataset. While this study was very focused and will provide useful insight, another automated classification problem is the BACH challenge which is ⁶⁰ to classify hematoxylin and eosin (H&E) stained breast biopsy slides and WSIs into distinct histologic categories (i.e., cancerous tumour subtypes).

The AI classifier achieved an accuracy on a par with a pathologist and improved accuracy from 80% to 88% (increase of 8%) and the average concordance between two pathologists improved from 83% to 90%. An example of an AI algorithm that was developed by a commercial provider is the GALEN breast algorithm. The GALEN algorithm ¹ utilized breast biopsy samples to detect breast cancer cells and classify those cells as per their histologic subtype. The values of the AUC obtained for the model on large data set collected by participants in the study showed that the AUC for detecting ¹²⁰ invasive carcinomas and ductal carcinoma in situ were 0.99 and 0.98, respectively (Ahn et al., 2023).

Precision Medicine

Breast cancer precision medicine represents the leading edge of innovative approaches to improve breast cancer patient outcomes through new approaches to diagnosing, treating and managing breast cancer. Precision medicine is changing how breast cancer is treated by focusing on the individual nature of each woman and tumor in order to develop a unique treatment plan versus the traditional "one-size-fits-all" approach of providing similar treatment to all patients, regardless of their individual characteristics.

While new advances in medical technology and early detection of breast cancer are constantly being made, more progress needs to be made regarding better results ¹² for breast cancer patients. Additionally, breast cancer is a very broad and diverse class of diseases making it difficult to come up with one specific treatment for each individual patient. This heterogeneity includes all of the different types of breast cancer, defined by unique DNA mutation patterns, molecular profiles, and cellular characteristics that

are each clinically significant in the experience of each individual with breast cancer, including his/her illness development and outcome of medication.

For example, the introduction of medications that specifically target specific types of breast cancer (e.g., HER2 inhibitors versus hormonal therapy) has created a significant improvement in measurable outcomes for women who have an HER2-positive or hormone-positive form of breast cancer by enabling oncologists to directly attack the abnormalities ² that drive the growth rate of their breast cancer. New treatment options using immunotherapy agents are being explored ² for patients with triple negative breast cancer, but further clinical testing is required to find out if these new treatments can be effective on all types of breast tumors.

Through the use of large-scale data analysis and AI, researchers can continuously learn more about the biology of breast tumor and develop personalized care plans. In addition, with the use of non-invasive liquid biopsy testing, cancer care teams can frequently check for real-time changes in the tumor, allowing for the opportunity to make timely adjustments to breast cancer treatment (Papalexis et al., 2024).

Novel Biomarkers

MicroRNAs as Novel Biomarkers ⁸ for Breast Cancer

Mi(cro)RNAs form a new group of small (< 25 nucleotides), non coding, ⁷³ endogenous RNA molecules. Since their introduction in 1993, mi(cro)RNAs have been implicated as important components in regulating many biological processes. Investigating the mechanisms by which mi(cro)RNAs operate is still in its infancy. Nonetheless, studies to date indicate that mi(cro)RNAs can regulate the expression of cellular genes through either transcriptional or post-transcriptional mechanisms, by inhibiting the translation of mRNA for protein coding genes or cleaving the mRNA and inducing its degradation as a result of imperfect base pairing between the mi(cro)RNA and its target mRNA [3]. It is reasonable to assume that miRNA biogenesis in human cells would be a complex process involving multiple steps. Figure 1 [4] presents a simplified illustration of how miRNA is created. For a miRNA to associate with its target mRNA, the specific area of the mature miRNA sequence that contains "seed" homology (i.e., base pairs 2 through 8) will be at the 5' end of the sequence. This area is commonly referred to as "seed sequence" [5]. Target identification algorithms have been created to find potential targets for mRNAs and place strong emphasis on the seed sequence, using it to locate

a complementary sequence in the 3' end of known genes based on conservation between groups of organisms (Heneghan, Miller, Lowery, Sweeney & Kerin, 2010).

New Targeted Therapies

Proceeding cautiously is important when making any claims about either the number of novel targets or the improvement of many models of anti-cancer treatments originating from improved knowledge about molecular mechanisms that cause cancer. Many new anti-cancer therapies (targeted therapies) work by trying to inhibit specific proteins in the signaling pathways responsible for the growth of the tumor. For example, many of the new targeted therapies under investigation as potential treatments for breast cancer are focused on inhibiting specific proteins in specific signaling pathways (e.g., proteins responsible for regulating cell growth, cellular adhesion and spreading, apoptosis, cell cycle regulation, and angiogenesis). The FDA in 2009 approved two breast cancer drugs: Lapatinib, which targets the HER2 receptor, and Bevacizumab, which targets the VEGF. Also, a number of other targeted treatments, such as the treatment of EGF receptor (EGFR), the combination of HER2 and EGFR, and VEGF-targeted drugs, are under development, some of which are already under clinical studies. Besides the development of targeted treatment for HER2 and VEGF, other types of signalings will also be examined within a clinical study, such as the signal pathway of RAS/MEK/ERK, PI3 kinase/Akt/mTOR, IGF/IGFR, PARP-1, and others (Alvarez et al, 2010).

Conclusion:

There is ample evidence that breast cancer is a serious problem of international public health significance as a result of its multiple risk factors and modes of manifestation. The goal behind this paper is to provide information about the epidemiology and the threat factors for breast cancer; including identifying high risk populations, and how both genetic factors and environmental factors play a role. In addition, the different types of breast cancer will be summarized with regard to their pathophysiology and molecular biology, highlighting the heterogeneity of the disease and the potential of personalized therapy. Thus, the mechanisms of breast tumor progression that are currently being explored through new research are outlined with a focus on complex pathways.

One interpretation of this review is that implementing previously described methods of prevention. For instance, the implementation of chemoprevention and lifestyle modifications would lead to considerable decrease in the risks associated with breast cancer. Second, taking into account the rate of advancement in cancer research, future advances in understanding the biology of breast cancer would likely result in development of novel preventive measures in the next ten years. However, one important challenge to applying what is known about effective means of prevention to appropriate populations of women has yet to be overcome. Given the process of chemotherapy prevention, we assume that there are systems in place to measure risk and evaluate the benefits and drawbacks of therapy and determine the appropriate use of medications.

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